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(https://swayam.gov.in) (https://swayam.gov.in/nc_details/NPTEL)

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NPTEL (https://swayam.gov.in/explorer?ncCode=NPTEL) » The Joy of Computing using Python (course)

Course outline

About NPTEL ()

How does an NPTEL online course work? ()

Week 1 ()

Week 2 ()

Week 3 ()

week 4 ()

Week 5 ()

Week 6 ()

Week 7 ()

Week 8 ()

Tuples- Python Data Structure (unit?unit=165&lesson=166)

Lottery Simulation - Profit or Loss (unit?

Week 8 : Assignment 8

The due date for submitting this assignment has passed.

Due on 2026-03-18, 23:59 IST.

Assignment submitted on 2026-03-18, 09:35 IST

A Month of Chance—Simulating a Lottery Strategy

Background

Ravi, a college student interested in probability and data analysis, wants to understand whether playing a simple lottery-style game every day can be profitable in the long run. Instead of relying on intuition or anecdotes, he decides to **simulate the experience using Python**.

Ravi chooses a very straightforward strategy:

- Every day, he selects **one fixed number between 1 and 10** as his bet.
- A system randomly draws a number between 1 and 10.
- If his chosen number matches the draw, he earns a large reward.
- If it doesn't, he incurs a small loss.

He repeats this experiment **once per day for 30 days**, representing a month of consistent participation.

Objective of the Study

The purpose of this case study is to:

- Observe how **randomness** affects outcomes over time
- Track **profit and loss trends** instead of isolated wins
- Visually analyze whether consistency in betting leads to gains or losses

Rather than focusing on a single lucky or unlucky day, Ravi wants to see the **overall trajectory** of his account balance.

Experimental Setup

unit=165&lesson=167)
Lottery Simulation - Profit or Loss - Part 01 (unit?unit=165&lesson=168)
Lottery Simulation - Profit or Loss - Part 02 (unit?unit=165&lesson=169)
Lottery Simulation - Profit or Loss - Part 03 (unit?unit=165&lesson=170)
Lottery Simulation - Profit or Loss - Part 04 (unit?unit=165&lesson=171)
Lottery Simulation - Profit or Loss - Part 05 (unit?unit=165&lesson=172)
Lottery Simulation - Profit or Loss - Part 06 (unit?unit=165&lesson=173)
Image Processing - Enhance your images (unit?unit=165&lesson=174)
Image Processing - Enhance your images - Part 01 (unit?unit=165&lesson=175)
Image Processing - Enhance your images - Part 02

- Ravi starts with an account balance of **₹0**
- He plays the game **once per day for 30 days**
- Each day's result updates his account balance
- The balance is recorded daily to study trends over time

To make the results more intuitive, Ravi plots:

- **Days (1–30)** on the x-axis
- **Account balance** on the y-axis

This allows him to clearly see fluctuations, streaks of loss, occasional jumps, and the final outcome.

Given Code for Analysis

Students are provided with the following Python code and are expected to analyze its behavior, output, and implications based on the case study above:

```
import random
import matplotlib.pyplot as plt

# User input
bet = int(input("Enter your bet from 1 to 10: "))

# Account initialization
account = 0

# Simulation loop for one month (30 days)
x = []  # Days
y = []  # Account balances

for day in range(30):
    lucky_draw = random.randint(1, 10)

    if bet == lucky_draw:
        account += 900
    else:
        account -= 100

    x.append(day + 1)
    y.append(account)

# Visualization
plt.plot(x, y)
plt.xlabel("Days")
plt.ylabel("Account Balance")
plt.title("Lottery Simulation - Profit or Loss")
plt.show()
```

1) What does the variable bet represent in the simulation? ***1 point***

- The amount of money invested each day
- The randomly generated lucky number
- The fixed number chosen by the user for the entire month
- The number of days the simulation runs

Yes, the answer is correct.
Score: 1

Accepted Answers:
The fixed number chosen by the user for the entire month

2) How many times is the lottery game simulated in the program? ***1 point***

(unit?unit=165&lesson=176)
Image Processing - Enhance your images - Part 03 (unit?unit=165&lesson=177)
Anagrams (unit?unit=165&lesson=178)
Anagrams - Part 01 (unit?unit=165&lesson=179)
Anagrams - Part 02 (unit?unit=165&lesson=180)
Anagrams - Part 03 (unit?unit=165&lesson=181)
Facebook Sentiment Analysis (unit?unit=165&lesson=182)
Facebook Sentiment Analysis - Part 01 (unit?unit=165&lesson=183)
Facebook Sentiment Analysis - Part 02 (unit?unit=165&lesson=184)
Facebook Sentiment Analysis - Part 03 (unit?unit=165&lesson=185)
Facebook Sentiment Analysis - Part 04 (unit?unit=165&lesson=186)

- 10
- 30
- Until the user wins
- Depends on the bet value

Yes, the answer is correct.
Score: 1

Accepted Answers:
30

3) What happens to the account balance when the bet does NOT match the lucky draw? ***1 point***

- It remains unchanged
- It increases by 100
- It decreases by 900
- It decreases by 100

Yes, the answer is correct.
Score: 1

Accepted Answers:
It decreases by 100

4) Which Python library is responsible for generating randomness in the simulation? ***1 point***

- matplotlib
- random
- pyplot
- math

Yes, the answer is correct.
Score: 1

Accepted Answers:
random

5) What do the lists x and y store respectively? ***1 point***

- Account balances and days
- Bets and lucky numbers
- Days and account balances
- Wins and losses

Yes, the answer is correct.
Score: 1

Accepted Answers:
Days and account balances

6) If the user never wins even once during the 30 days, what will be the final account balance? ***1 point***

- −3000
- −900
- −100
- 0

Yes, the answer is correct.

- **Quiz: Week 8 : Assignment 8 (assessment?name=658)**
- Week 8 : Programming Assignment 1 (/noc26_cs84/progassignment?name=635)
- Week 8 : Programming Assignment 2 (/noc26_cs84/progassignment?name=636)
- Week 8 : Programming Assignment 3 (/noc26_cs84/progassignment?name=638)
- Week 8 Feedback Form: The Joy of Computing using Python (unit?unit=165&lesson=187)

Week 9 ()

Week 10 ()

Week 11 ()

Week 12 ()

Text Transcripts ()

Download Videos ()

Score: 1

Accepted Answers:
−3000

7) What is the probability of winning on any single day? ***1 point***

- 1/30
- 1/100
- 1/10
- Depends on previous days

Yes, the answer is correct.
Score: 1

Accepted Answers:
1/10

8) Why does the account balance graph typically show long downward trends with occasional sharp upward jumps? ***1 point***

- Because the bet value changes daily
- Because losses are frequent and wins are rare but large
- Because matplotlib smooths the curve
- Because account balance is averaged

Yes, the answer is correct.
Score: 1

Accepted Answers:
Because losses are frequent and wins are rare but large

9) Which change would make the simulation more statistically reliable without changing the game rules? ***1 point***

- Increasing the win amount
- Running the simulation for more days
- Changing the bet every day
- Removing the plot

Yes, the answer is correct.
Score: 1

Accepted Answers:
Running the simulation for more days

10) Despite a high winning reward (₹900), why does the simulation still tend to show losses over time? ***1 point***

- Because the plotting scale is incorrect
- Because the probability of winning is too low compared to the loss frequency
- Because Python integers overflow
- Because the account starts at zero

Yes, the answer is correct.
Score: 1

Accepted Answers:
Because the probability of winning is too low compared to the loss frequency

Detecting Pattern Consistency Between Two Strings

Background

A software team working on a **text pattern–matching system** needs to verify whether two given strings follow the **same structural pattern**. This requirement arises in areas such as:

- Data normalization
- Search indexing
- Language processing
- Pattern validation in user input

The focus is **not** on object-oriented design, but purely on **string processing and logical consistency**.

Problem Scenario

Two strings are given:

- String **S** represents an original pattern
- String **T** represents a transformed version

The task is to determine whether:

- Each character in **S** consistently maps to **one unique character** in **T**
- The mapping is **one-to-one and reversible**
- The relative structure of both strings remains identical

For example:

- "egg" and "add" follow the same pattern
- "foo" and "bar" do not

Constraints of the Study

- Both strings must be of **equal length**
- A character cannot map to **multiple different characters**
- Two different characters cannot map to the **same character**
- The comparison is performed **position by position**

Learners are given the following code to analyze and answer questions based on the case study:

```
def isomorphicString(s, t):
    if len(s) != len(t):
        return False

    s_map = {}
    t_map = {}

    for i in range(len(s)):
        char_s = s[i]
        char_t = t[i]

        if (char_s in s_map and s_map[char_s] != char_t) or
           (char_t in t_map and t_map[char_t] != char_s):
            return False

        s_map[char_s] = char_t
        t_map[char_t] = char_s

    return True
```

11) What will the function return for s = "egg" and t = "add"? ***1 point***

- (selected) True
- False
- Depends on dictionary insertion order
- Raises an error

Yes, the answer is correct.
Score: 1

Accepted Answers:
True

12) What is the first condition that can cause the function to return False? ***1 point***

- Duplicate characters in s
- Conflicting mappings in dictionaries
- Repeated characters at adjacent positions
- (selected) Length mismatch between the two strings

Yes, the answer is correct.
Score: 1

Accepted Answers:
Length mismatch between the two strings

13) What will the function return for s = "ab" and t = "cc"? ***1 point***

- (selected) False
- True
- Only the forward mapping fails
- Depends on loop iteration

Yes, the answer is correct.
Score: 1

Accepted Answers:
False

14) Why is the condition checking **both** s_map and t_map necessary? ***1 point***

- To reduce memory usage
- To avoid nested loops
- To speed up execution
- (selected) To enforce a one-to-one and reversible character mapping

Yes, the answer is correct.
Score: 1

Accepted Answers:
To enforce a one-to-one and reversible character mapping

15) Which of the following best describes what the algorithm is verifying? ***1 point***

- (selected) Structural equivalence between two strings
- Lexicographical ordering
- Equality of character frequencies
- Whether both strings are permutations

Yes, the answer is correct.

Score: 1

Accepted Answers:

Structural equivalence between two strings